

Ethan Buckley

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Education

University College London, MSci Natural Sciences

Sep 2024 to Jun 2028

- Physics major, Physical Chemistry minor.
- Relevant modules: Mathematical Methods III, Quantum Physics, Thermal and Statistical Physics.
- Year 3 literature review (in progress), supervised by Professor Peter Coveney: hybrid quantum and GPU computing for biomolecular simulation, including variational quantum algorithms.

Experience

Data Science and Research Intern, Down Syndrome Education International

Jun 2026 to Sep 2026

- Built hierarchical Bayesian models in PyMC for vocabulary growth in children with Down syndrome. Fitted 1,424 assessments from 14 datasets in six countries, with Gaussian-process age trends and random effects for each study and child.
- Developed intention-to-treat, difference-in-differences and mediation models for a Bayesian re-analysis of a randomised reading trial (54 children, four assessment waves), following the study's causal diagram. Ranked predictors with LightGBM and SHAP.
- Ran prior and posterior predictive checks, compared models by PSIS-LOO and leave-one-study-out cross-validation, and flagged any fit with $\hat{R}$ above 1.01 or effective sample size below 400.
- Compared text-to-speech voices from four providers through blind A/B listening and Bradley-Terry ranking. Fixed a scoring bug that gave tied comparisons to whichever voice sorted first.

AI Training Specialist (freelance), DataAnnotation

Jun 2026 to Aug 2026

- Evaluated large language model answers in physics, chemistry, maths and machine learning, ran any code they contained, wrote reference solutions and produced comparison data.

Seasonal Team Member, Roadchef, Killington Lake

Jul 2025 to Apr 2026

- Worked during university holidays, cleaning the motorway services site, receiving cleaning-supply deliveries and covering shifts at Costa, WHSmith and Days Inn.

Work experience, BAE Systems (Barrow) and EDF Energy (Heysham 1)

Jul 2023 to Aug 2023

- Spent three days with engineering teams at Barrow and five days with the chemistry and maintenance departments at Heysham 1, including observing reactor-chemistry monitoring.

Projects

- **SwimSignal (independent):** Built a pollution-risk forecasting service combining rainfall, live storm-overflow feeds and river-network data. Trained a gradient-boosted spill model with per-overflow calibration. On 826,725 held-out United Utilities overflow-days in 2025, its Brier score was a third lower than site climatology (0.0445 against 0.0669). Published forecasts and verification at swimsignal.co.uk.
- **S&P 500 stock screener (independent):** Built an XGBoost classifier with 22 features and FinBERT news sentiment. Walk-forward validation over 1,659 out-of-sample days (2020 to 2026) gave about 46% top-15 precision against a 29% base rate, matching a volatility-only baseline. The portfolio trailed the S&P 500 after costs, with some survivorship bias remaining.

Skills

- **Programming:** Python (NumPy, pandas, PyMC, XGBoost, LightGBM, SHAP, scikit-learn, pytest), SQL and DuckDB, Git and GitHub Actions, Quarto, LaTeX, generative-model batch APIs.
- **Statistics and machine learning:** Bayesian inference and MCMC, hierarchical models, Gaussian processes, gradient boosting, causal inference (DAGs, intention-to-treat, difference-in-differences, mediation), blind A/B evaluation with Bradley-Terry ranking, NLP with FinBERT.

Activities and interests

- Player, UCL Men's Hockey Club; volunteer junior coach, Kirkby Lonsdale Hockey Club.
- UCL AI Society and Natural Sciences Society.
- Interests: mountaineering, skiing, running.